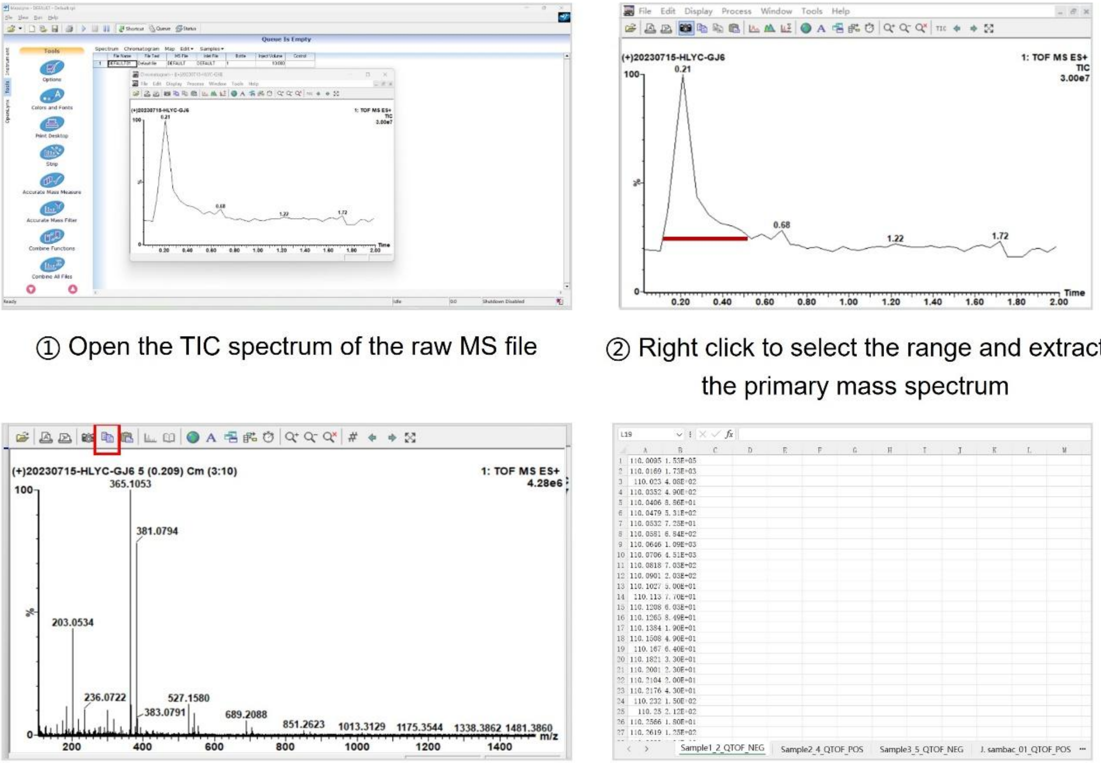
MS Raw Data Conversion

1.1 HT-MS data extraction in Waters system by MassLynx

MassLynx software was used to extract data from Waters DI-QTOF MS acquisitions, generating data list files in .xls, .xlsx, or .csv formats containing *m/z* values and their corresponding intensities. The specific process is as follows:



① Open the TIC spectrum of the raw MS file

② Right click to select the range and extract the primary mass spectrum

③ Export raw data

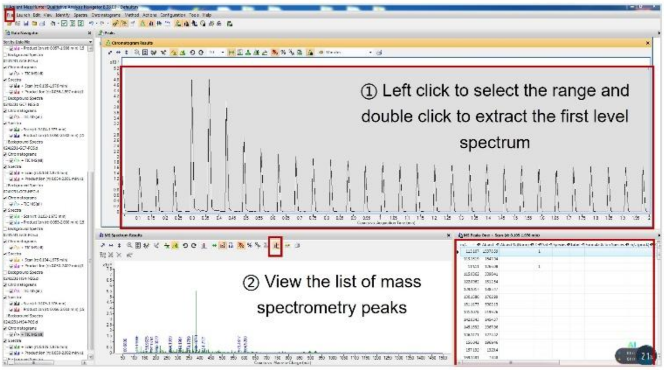
④ Paste into an EXCEL spreadsheet

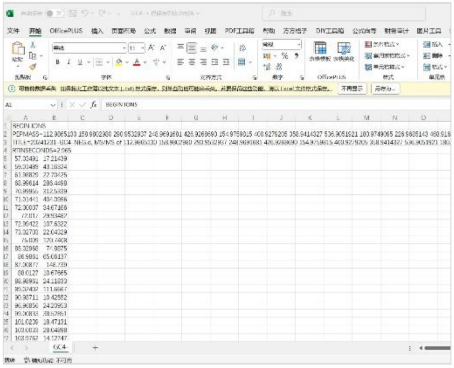
Figure S1-1 HS-MS data extraction process using Masslynx

1.2 HT-MS data extraction in Agilent system by MassHunter

MassHunter software was utilized to extract MS data from Agilent DI-QTOF acquisitions, generating data list files in .xls, .xlsx, or .csv formats containing *m/z* values and their corresponding intensities. These files were subsequently imported into RapidMass for data processing, and the specific process is as follows:

③ Export MGF file



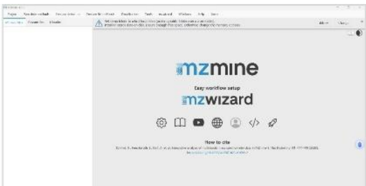


④ Open the mgf file and save it as an Excel file(xlsx.)

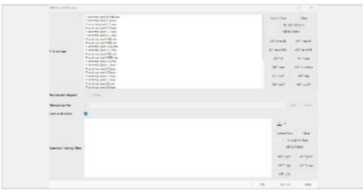
Figure S1-2 HS-MS data extraction process using Masshunter

1.3 MS data extraction in Thermo system and other systems by MZmine

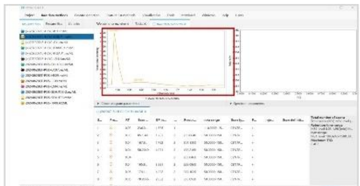
MZmine software was employed to extract MS data from Thermo DI-Orbitrap acquisitions and DART-MSdata from literature, producing data list files in .xls, .xlsx, or .csv formats containing *m/z* values and their corresponding intensities. Additionally, a "Multi-System Data Processing Guide" has been added to the user manual, which provides instructions on converting raw MS data into a universal format and offers solutions and recommendations for potential issues encountered during data processing across different systems. The complete user manual is available for download at https://github.com/Katherine00689.



① Download software, register an account

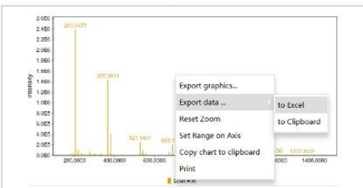


② Open the interface for importing raw data files

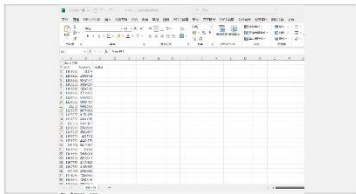


③ Select TIC spectrum, double-click to extract primary mass spectrum

④ Select the mass spectrum for extracting the main peak



⑤Export primary spectral data



⑥ The data format is a binary list of mz and intensity

Figure S1-3 HS-MS data extraction process using Mzmine